INCH-POUND

MIL-DTL-28840/5C
13 May 2009
SUPERSEDING
MIL-DTL-28840/5B
20 April 2004

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, STRAIGHT, METAL CONDUIT FOR EMI SHIELDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

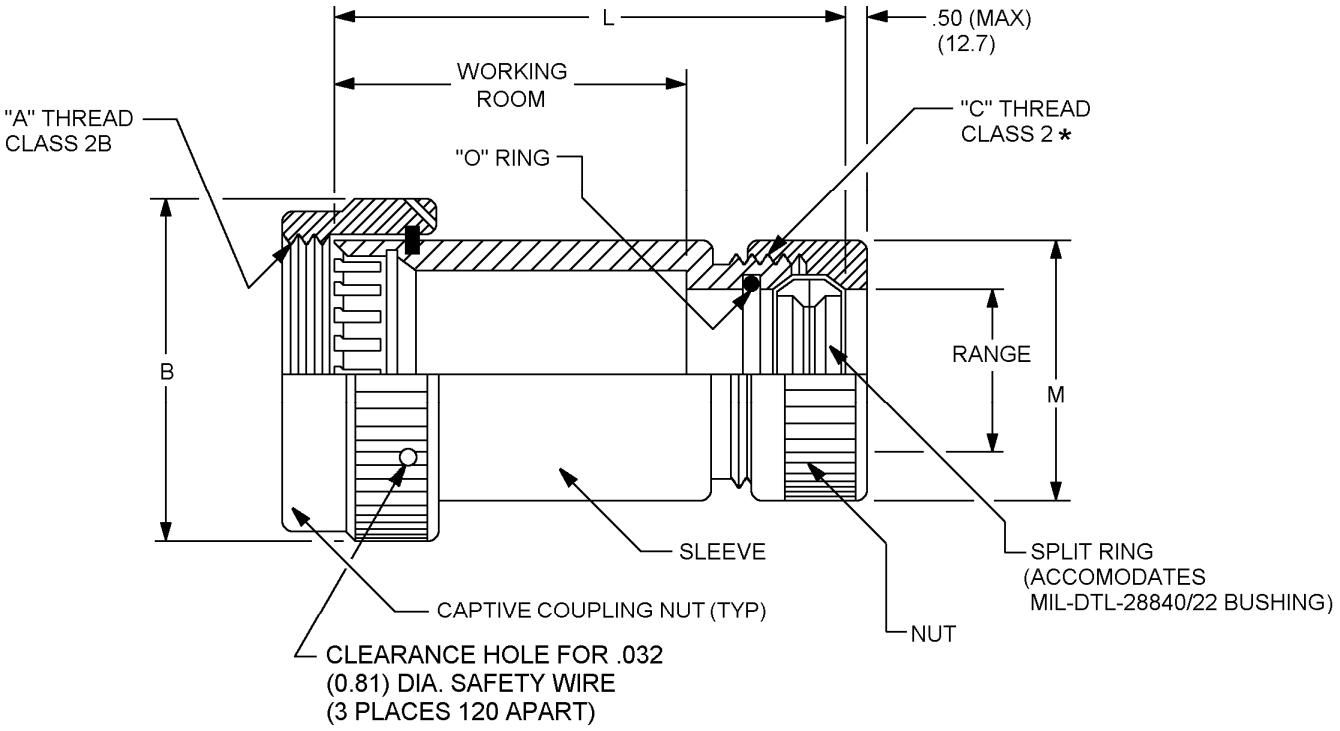


FIGURE 1. Dimensions and configurations.

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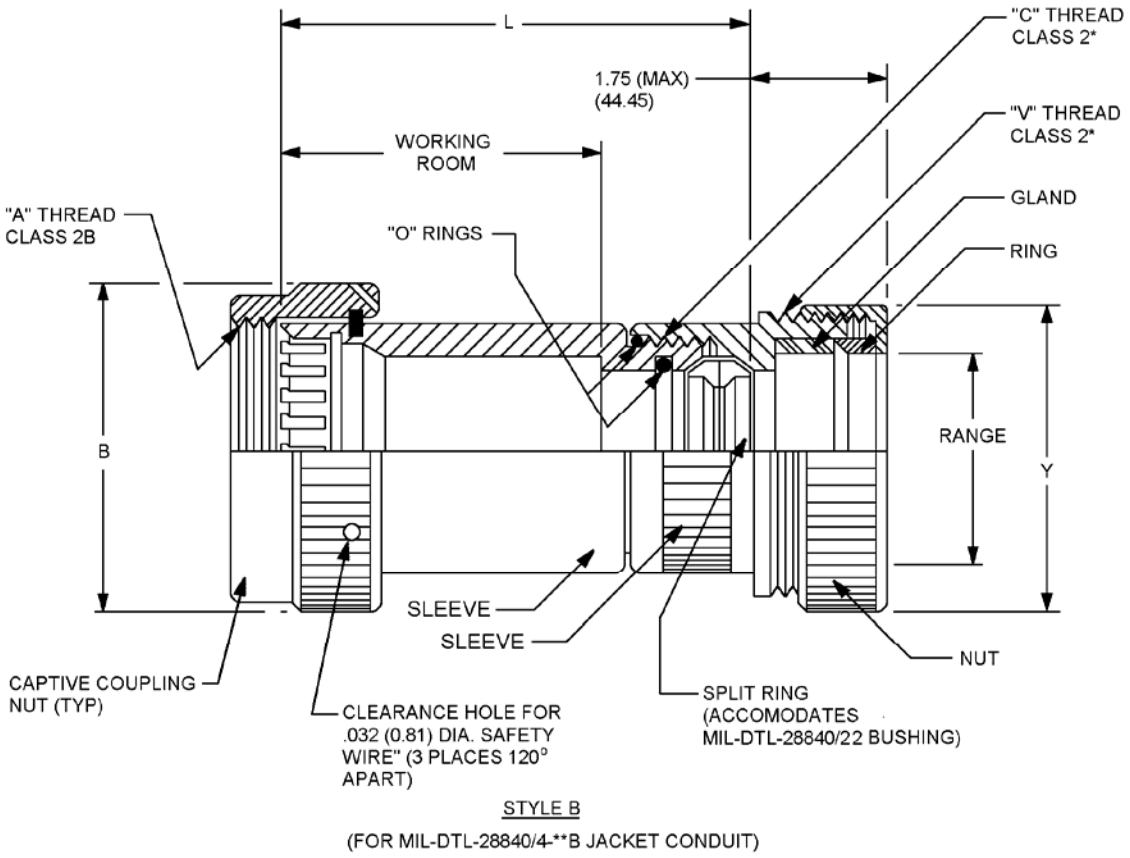


FIGURE 1. Dimensions and configurations - Continued.

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Dash no.	Designator (see note 1)	A thread class 2B	B max dia	C thread, class 2* (see note 5)	Sealing range	
					Max	Min
01	A (11)	.750-20 UNEF	1.028 (26.11)	.875-20 UNEF	.63 (16.0)	.50 (12.7)
02	B (13)	.875-20 UNEF	1.141 (28.98)			
03				1.000-20 UNEF	.75 (19.1)	.63 (16.0)
04	C (15)	1.000-20 UNEF	1.263 (32.08)	.875-20 UNEF	.63 (16.0)	.50 (12.7)
05				1.000-20 UNEF	.75 (19.1)	.63 (16.0)
06				1.125-18 UNEF	.88 (22.4)	.75 (19.1)
07				1.250-18 UNEF	1.00 (25.4)	.88 (22.4)
08	D (17)	1.125-18 UNEF	1.387 (35.23)	1.125-18 UNEF	.88 (22.4)	.75 (19.1)
09				1.250-18 UNEF	1.00 (25.4)	.88 (22.4)
10				1.625-18 UNEF	1.38 (35.1)	1.19 (30.2)
11	E (19)	1.250-18 UNEF	1.513 (38.43)	1.125-18 UNEF	.88 (22.4)	.75 (19.1)
12				1.250-18 UNEF	1.00 (25.4)	.88 (22.4)
13				1.625-18 UNEF	1.38 (35.1)	1.19 (30.2)
14	F (23)	1.438-18 UNEF	1.703 (45.51)	1.250-18 UNEF	1.00 (25.4)	.88 (22.4)
15				1.625-18 UNEF	1.38 (35.1)	1.19 (30.2)
16				1.875-18 UNS	1.63 (41.4)	1.44 (36.6)
17	G (25)	1.562-18 UNEF	1.825 (46.35)	1.250-18 UNEF	1.00 (25.4)	.88 (22.4)
18				1.625-18 UNEF	1.38 (35.1)	1.19 (30.2)
19				1.875-18 UNS	1.63 (41.4)	1.44 (36.6)
20				2.125-16 UN	1.88 (47.8)	1.63 (41.4)
21	H (29)	1.875-16 UN	2.143 (54.43)	1.875-18 UNS	1.63 (41.4)	1.44 (36.6)
22				2.125-16 UN	1.88 (67.8)	1.63 (41.4)
23	J (33)	2.062-16 UNS	2.329 (59.16)	2.125-16 UN	1.88 (47.8)	1.63 (41.4)
24				2.750-16 UN	2.38 (60.5)	2.13 (54.1)
25	A (11)	.750-20 UNEF	1.028 (26.11)	.875-20 UNEF	.75 (19.05)	.63 (16.00)

FIGURE 1. Dimensions and configurations - Continued.

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Dash no.	Designator (see note 1)	Conduit size	L ± .12 (± 3.05)	Working room ref.	V thread class 2* (see note 5)	M max dia	Y max dia
01	A (11)	03	1.75 (44.45)	1.00 (25.4)	1.000-20 UNEF	1.12 (28.45)	1.19 (30.23)
02	B (13)						
03		04			1.188-18 UNEF	1.25 (31.8)	1.38 (35.1)
04	C (15)	03			1.000-20 UNEF	1.12 (28.5)	1.19 (30.23)
05		04			1.188-30 UNEF	1.25 (31.8)	1.38 (35.1)
06		05			1.438-18 UNEF	1.43 (36.3)	1.63 (41.4)
07		06	3.00 (76.2)		1.750-18 UNS	1.62 (41.2)	1.94 (49.3)
08	D (17)	05	2.25 (57.15)	1.50 (38.10)	1.438-18 UNEF	1.43 (36.3)	1.63 (41.4)
09		06			1.750-18 UNS	1. 62 (41.2)	1.94 (49.3)
10		08		2.00 (50.80)	2.00-18 UNS	1.81 (46.0)	2.19 (55.6)
11	(E19)	05		1.50 (38.10)	1.438-18 UNEF	1.43 (36.3)	1.63 (41.4)
12		06			1.750-18 UNEF	1.62 (41.2)	1.94 (49.3)
13		08			2.00-18 UNS	1.81 (46.2)	2.19 (55.6)
14	F (23)	06	2.75 (69.85)	2.00 (50.80)	1.750-18 UNS	1.62 (41.2)	1.94 (49.3)
15		08			2.00-18 UN	1.81 (46.0)	2.19 (55.6)
16		10	3.50 (88.90)		2.00-16 UN	2.18 (55.4)	2.44 (62.0)
17	G (25)	06	2.75 (60.85)		1.750-18 UNS	1.62 (41.2)	1.94 (49.3)
18		08			2.00-18 UN	1.81 (46.0)	2.19 (55.6)
19		10			2.250-16 UN	2.18 (55.4)	2.44 (62.0)
20		12	4.00 (101.6)	2.50 (63.50)	2.500-16 UN	2.56 (65.0)	2.69 (68.3)
21	H (29)	10	3.25 (82.55)		2.250-16 UN	2.18 (55.4)	2.44 (62.0)
22		12			2.500-16 UN	2.56 (63.0)	2.66 (68.3)
23	J (33)	12				2.56 (65.0)	2.60 (88.3)
24		16			3.00-16 UN	3.06 (77.8)	3.18 (81.0)
25	A (11)	04	1.75 (44.45)	1.00 (25.4)	1.188-18 UNEF	1.75 (31.75)	1.38 (95.05)

NOTES:
1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches. Dimensions apply after plating.
3. Metric equivalents are given for information only.
4. Dimensions apply after plating.
5. See MIL-DTL-28840 for C and V thread, class 2*.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS

Dimensions and configurations: See figure 1.

Interface dimensions in accordance with MIL-DTL-28840.

Material: Aluminum alloy in accordance with ASTM B85, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211, or ASTM B221. Corrosion resistant steel in accordance with ASTM A276, ASTM A473, or ASTM A580/A580M, type 316.

Finish: Aluminum components finish shall be in accordance with MIL-DTL-28840.

Corrosion resistant steel components shall be cadmium plate, black.
The screws and washers shall be passivate in accordance with SAE-AMS2700, type 2.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the PIN.

PIN example:

M28840/ 5 16 W A

Military designator

Specification sheet number

Dash number

Material and finish designator
- W – Aluminum, cadmium plate, olive drab over electroless nickel
- D – Corrosion resistant steel, cadmium plate, black
- F – Wrought aluminum, pure electrodeposited aluminum – non-reflective
- L - Wrought aluminum, Nickel Fluorocarbon Polymer – non-reflective
- S - Wrought aluminum, zinc nickel – non-reflective

Style designator
- A – EMI/Non-environmental
- B – EMI/Environmental

Qualification: Two samples in each shell size for which qualification is desired, shall be tested in accordance with MIL-DTL-28840.

Qualification of MIL-DTL-28840/6 will qualify MIL-DTL-28840/5 by similarity for group 5 tests.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Referenced documents. In addition to MIL-DTL-28840, this document references the following:

MIL-DTL-28840/6	ASTM B108/B108M
MIL-DTL-28840/22	ASTM B209
ASTM A276	ASTM B211
ASTM A473	ASTM B221
ASTM A580/A580M	SAE-AMS2700
ASTM B26/B26M	
ASTM B85	

CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2007-217)

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